What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Aortic Stenosis

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Adapted from: Aortic Stenosis

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of aortic stenosis (AS).
2. Explain the pathophysiology of AS.
3. Interpret clinical findings to properly diagnose AS.
4. Outline the management plan of AS.

CASE CONNECTION

"I can't believe it. They turned me down for life insurance. Me, a healthy guy!" TR, a 46-year-old male, tells his wife while reading the letter from the insurance company. "They say my ECG is abnormal." TR makes an appointment to get a physical. The ECG he brings shows changes consistent with left ventricular hypertrophy. On exam, TR's blood pressure is 152/88 mm Hg. A crescendo-decrescendo systolic murmur can be heard all the way to the carotids. You know that this physical exam finding makes the diagnosis.

What is the murmur? How would you explain the diagnosis to TR? Consider your answers as you read, and we'll revisit TR at the end of the brick.

GO TO CONCLUSION ↓

What Is Aortic Stenosis?

Aortic stenosis (AS) is a condition in which the aortic valve doesn't open completely, meaning the heart has to work harder to pump blood past the narrowed opening to deliver blood to the rest of the body.

Anatomy

The aortic valve is positioned between the left ventricle (LV) and the aorta. When the LV contracts (systole), the aortic valve leaflets open, letting blood flow out of the LV and into the rest of the body. In AS, the aortic valve leaflets become stiff and unable to fully open, making it difficult for the heart to pump blood out to the rest of the body (Figure 1).

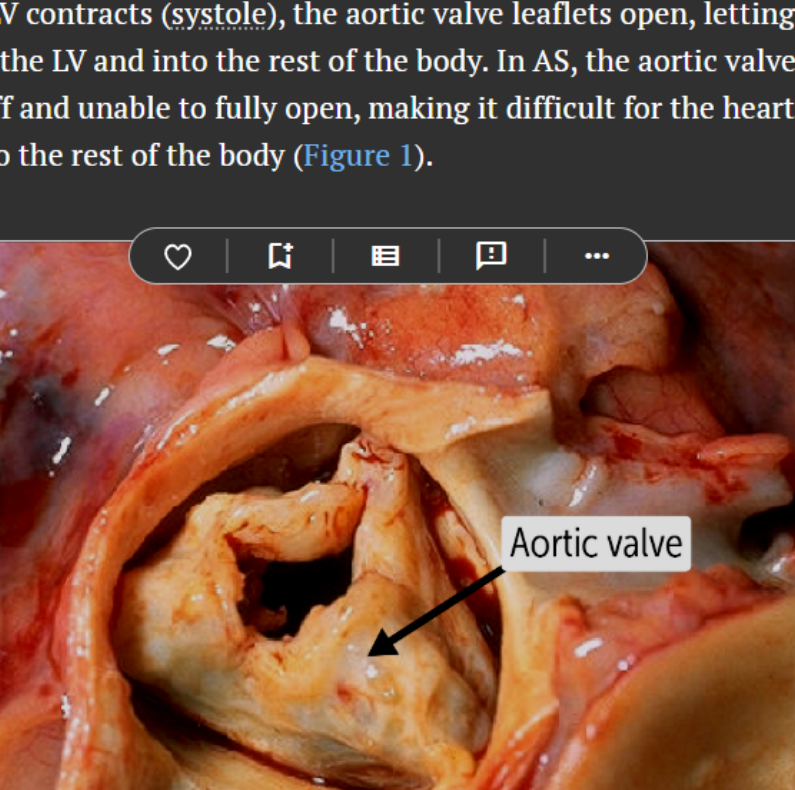

Figure 1. Gross Pathology of Aortic Stenosis.

When the valve leaflets can't open completely, blood flow out of the LV during systole is reduced. If the restriction of blood from the LV becomes severe enough, it can lead to a lack of oxygen getting to the various tissues in the body (e.g., brain, vital organs, muscles, etc). This deficit between the demands of the body and the heart's ability to pump blood well enough to keep pace is known as heart failure (HF).

Why is blood flow reduced during systole in aortic stenosis?

Epidemiology

AS is commonly seen in patients age >65 years old. Most cases in which the patient is over the age of 75 are a result of calcification and occur more commonly in the male population. The prevalence of AS in the older adult population is about 2%-9%.

How Do Patients With Aortic Stenosis (AS) Present? (LO1)

Patients with AS can be asymptomatic for years until the AS progresses to a severe point. Because patients are often asymptomatic, it's likely that AS will be discovered on a routine exam or during a check for another condition, when the clinician hears the characteristic murmur.

After patients begin to experience symptoms, their survival starts to decrease. The three classic sym rmation about a patient's prognosis—angina, syncope, and dyspnea.

Angina

Chest pain is caused by decreased blood flow through the coronary arteries to the myocardium (heart muscle). Angina occurs in AS because the coronary arteries branch from the aorta immediately distal to the aortic valve, at the very start of the ascending aorta. Decreased flow from the LV through the stenotic valve then leads to decreased coronary blood flow and angina. Patients with angina have an average survival of only 5 years from the onset of symptoms unless the valve disease is corrected.

Syncope

Loss of consciousness is due to the transient decrease of blood flow to the brain, again because of the tight stenotic valve. These patients have an average survival of 3 years from the onset of symptoms.

Dyspnea

Shortness of breath, a symptom of heart failure, occurs because the pressure on the left ventricle eventually causes it to hypertrophy and then causes the ventricular muscle to weaken. This leads to backup of fluid into the lungs, causing dyspnea due to pulmonary edema. These patients have an average survival of 2 years from the onset of symptoms.

How does aortic stenosis classically present?

What Is the Pathophysiology of Aortic Stenosis? (LO 2)

Most of the time, the aortic valve leaflets become stiff and immobile because of calcification of the valve. Less common causes include a congenital bicuspid valve (patients are born with two valve leaflets instead of the usual three) and rheumatic heart disease.

Valve Calcification

Over time, inflammation causes the aortic valve to scar, thicken, and eventually calcify and stiffen. Risk factors include:

- Older age
- Male sex
- Smoking
- Hypertension
- High levels of low-density lipoprotein (LDL) cholesterol
- Diabetes mellitus

This calcification is shown in Figure 2.

What are the risk factors that lead to aortic sclerosis?

Congenital Bicuspid Valve

A bicuspid aortic valve is the most common congenital malformation of the aortic valve and occurs in 1% to 2% of the population. Figure 3 shows a normal tricuspid aortic valve (left) and a bicuspid aortic valve (right).

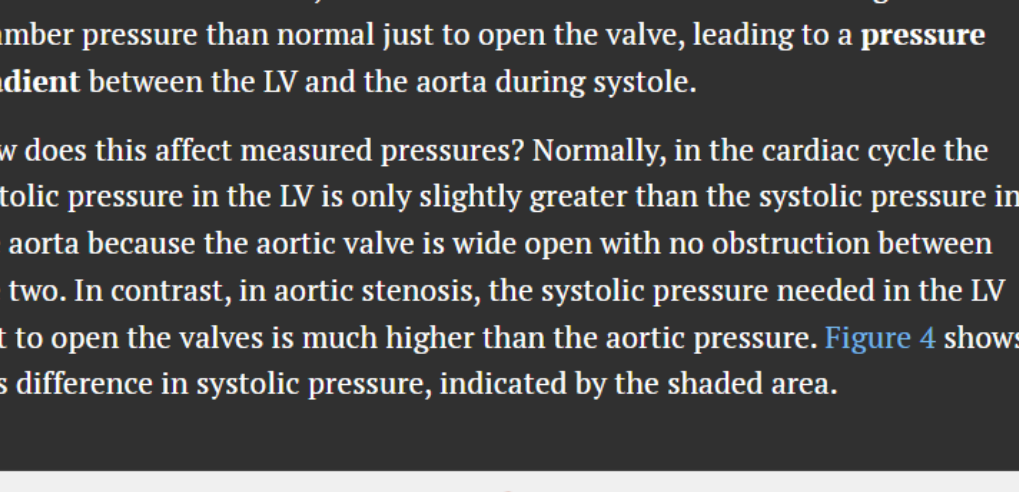

Figure 3. Normal and Bicuspid Aortic Valves.

left untreated, the body creates antibodies to a surface protein on the bacteria (M protein) that mimics a protein on the lining of the heart. This in turn directs the body's immune system to attack the lining of the heart, causing inflammation and scarring of the heart valves, most commonly the mitral and aortic valves. This scarring leads to calcification and stenosis of one or both valves.

Rheumatic heart disease is mostly seen in developing countries where infections are sometimes not treated as early as in countries with more developed medical care.

Physiology

AS causes the left ventricle (LV) to work much harder. Instead of pumping blood through a wide-open valve, the LV must pump through a stiffer, narrowed valve. To do this, the contraction of the LV must create higher chamber pressure than normal just to open the valve, leading to a **pressure gradient** between the LV and the aorta during systole.

How does this affect measured pressures? Normally, in the cardiac cycle the systolic pressure in the LV is only slightly greater than the systolic pressure in the aorta because the aortic valve is wide open with no obstruction between the two. In contrast, in aortic stenosis, the systolic pressure needed in the LV just to open the valves is much higher than the aortic pressure. Figure 4 shows this difference in systolic pressure, indicated by the shaded area.

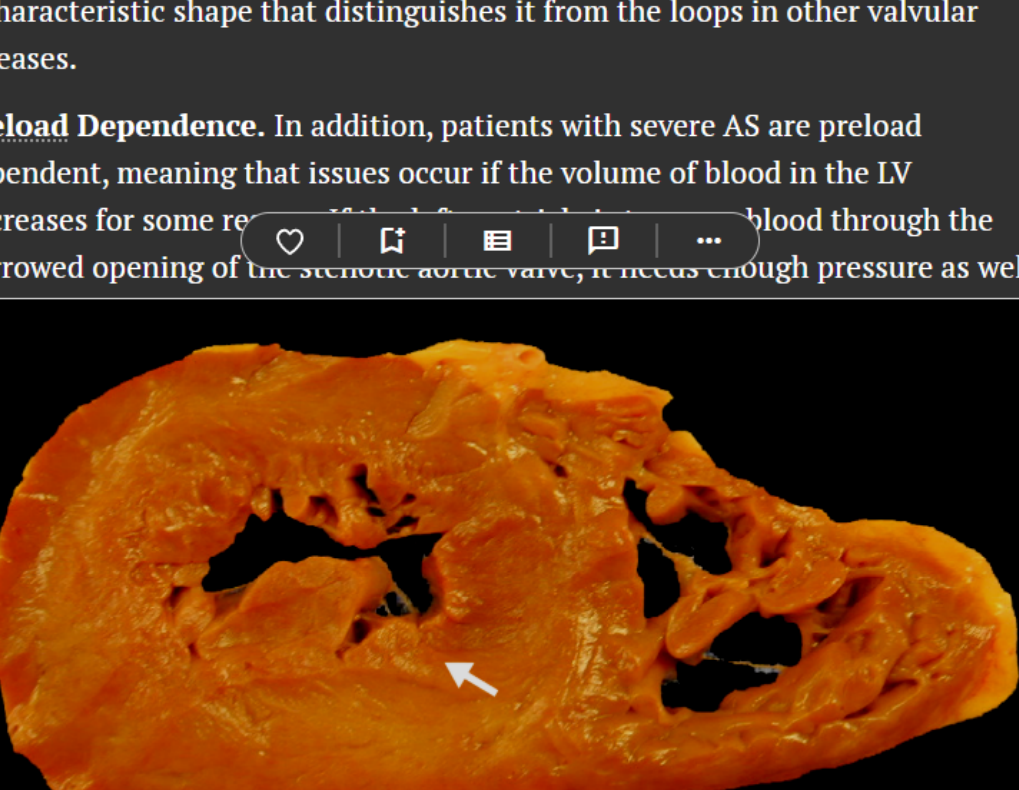

Figure 5. Pressure Volume Loop of Aortic Stenosis.

Afterload. The increase in the pressure gradient increases the **afterload** on the left ventricle, like what is seen in patients with very high systemic blood pressure. Recall that an increased afterload reduces the amount of ventricular force of contraction devoted to blood movement vs pressure generation, leading to increased end-systolic volume. This reduces the stroke volume (Figure 5, red arrow). Note that the pressure-volume loop in AS is tall and thin, a characteristic shape that distinguishes it from the loops in other valvular diseases.

Preload Dependence. In addition, patients with severe AS are preload dependent, meaning that issues occur if the volume of blood in the LV decreases for some re blood through the narrowed opening of the stenotic aortic valve, it needs enough pressure as well

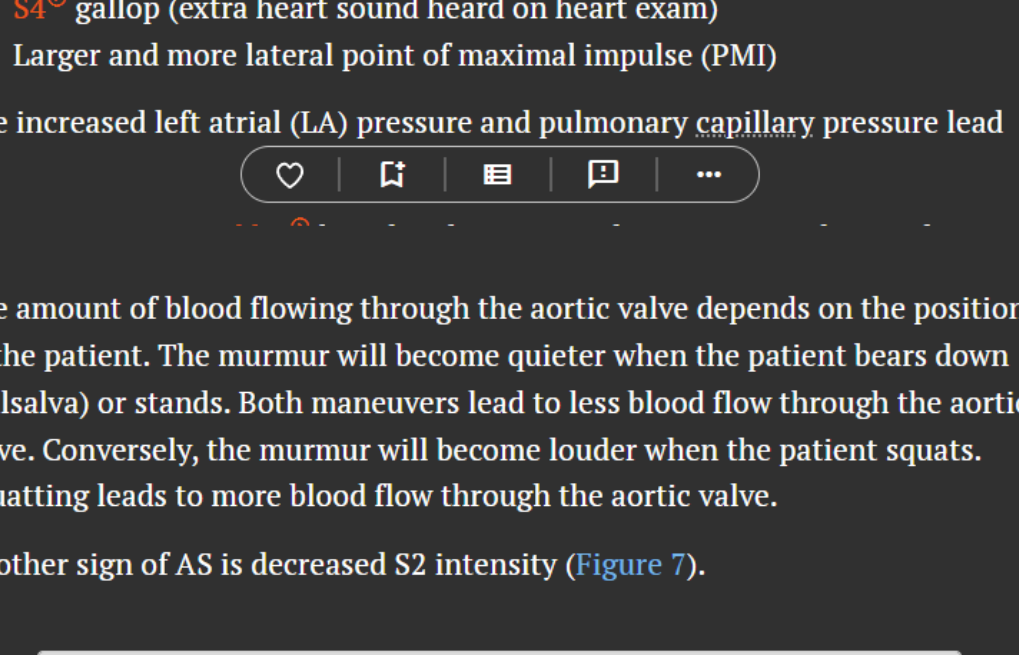

Figure 6. Hypertrophied Left Ventricle.

The thickened LV (arrow) causes a decrease in ventricular compliance, which leads to **diastolic dysfunction** (nonsystolic HF). In diastolic dysfunction, the thickened ventricle cannot fill well, decreasing the LV end-diastolic volume and again decreasing stroke volume.

As it worsens, AS can also cause other typical signs of nonsystolic HF. The hypertrophied ventricle leads to:

- S4 gallop (extra heart sound heard on heart exam)
- Larger and more lateral point of maximal impulse (PMI)

The increased left atrial (LA) pressure and pulmonary capillary pressure lead to:

The amount of blood flowing through the aortic valve depends on the position of the patient. The murmur will become quieter when the patient bears down (Valsalva) or stands. Both maneuvers lead to less blood flow through the aortic valve. Conversely, the murmur will become louder when the patient squats. Squatting leads to more blood flow through the aortic valve.

Another sign of AS is decreased S2 intensity (Figure 7).

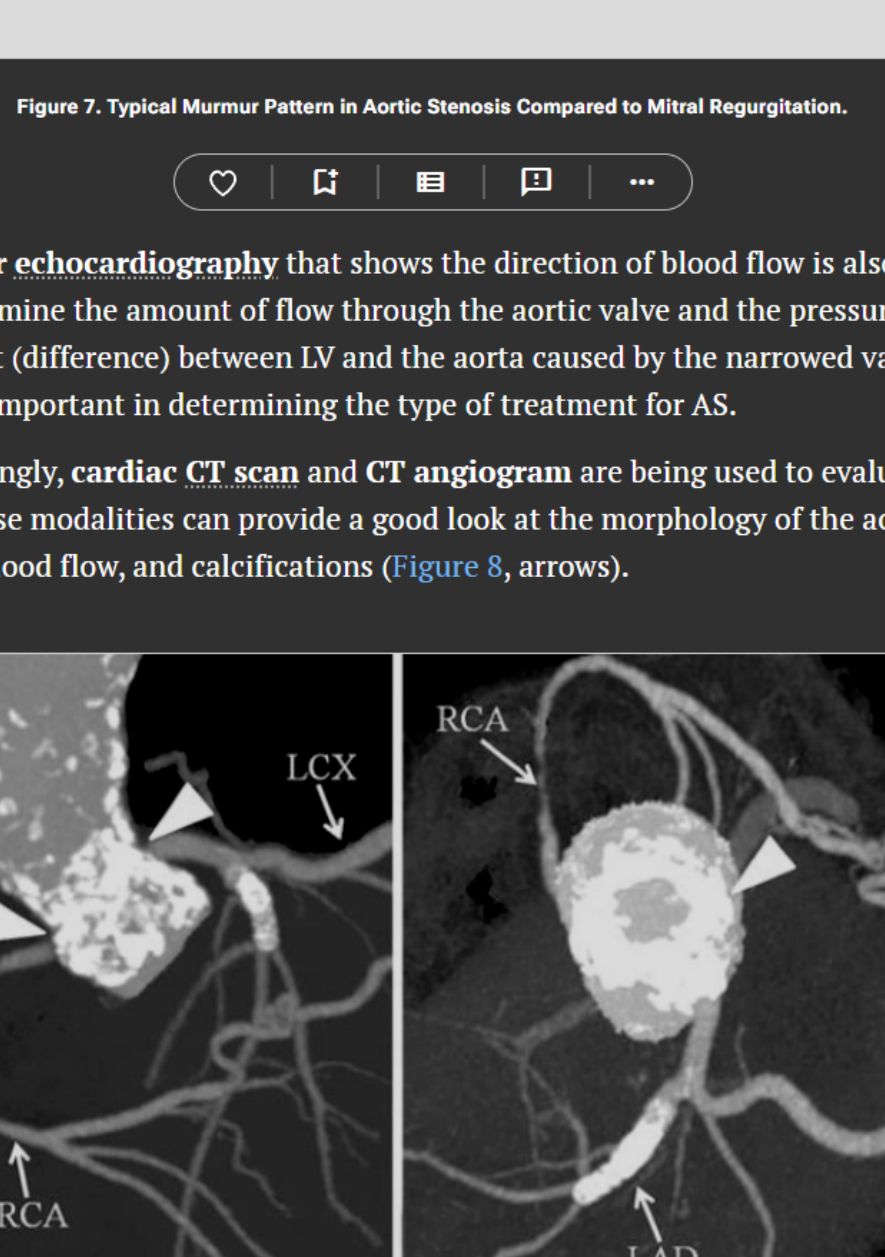

Figure 7. Typical Murmur Pattern in Aortic Stenosis Compared to Mitral Regurgitation.

Doppler echocardiography that shows the direction of blood flow is also used to determine the amount of flow through the aortic valve and the pressure gradient (difference) between LV and the aorta caused by the narrowed valve. This is important in determining the type of treatment for AS.

Increasingly, **cardiac CT scan** and **CT angiogram** are being used to evaluate AS. These modalities can provide a good look at the morphology of the aortic valve, blood flow, and calcifications (Figure 8, arrows).

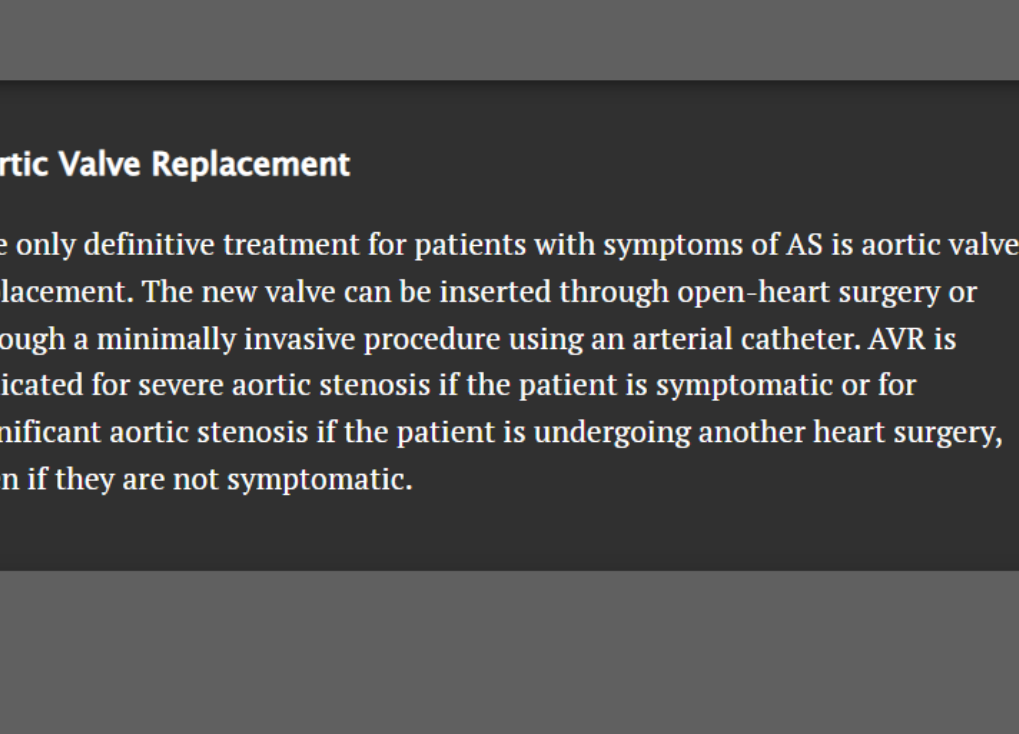

Figure 8. Severely Calcified Aortic Root on Cardiac CT Scan.

Arteries shown in Figure 8 include the right coronary artery (RCA), the

Why is nitroglycerin (NTG) harmful to many patients with AS?

Aortic Valve Replacement

The only definitive treatment for patients with symptoms of AS is aortic valve replacement. The new valve can be inserted through open-heart surgery or through a minimally invasive procedure using an arterial catheter. AVR is indicated for severe aortic stenosis if the patient is symptomatic or for significant aortic stenosis if the patient is undergoing another heart surgery, even if they are not symptomatic.

Which patients should undergo aortic valve replacement (AVR)?

years after HF.

Mnemonic

Symptoms of AS and associated average survival: **ASD 5-3-2** = **A**ngina, **s**yncope, **d**yspnea: **5** years, **3** years, **2** years.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to TR, what is the murmur? How do you explain the diagnosis to him?

TR's physical examination is consistent with aortic valve stenosis. Based on his age, the likely etiology is a bicuspid aortic valve. You explain that his heart works harder to push out the blood through the narrowed valve, and with this extra work, his heart muscle, like any other muscle, gets bigger. TR has an echocardiogram with Doppler, and because he is asymptomatic, he can be followed up with serial echocardiograms to assess any progression of the stenosis. He is told to contact you immediately if he has chest pain, shortness of breath, or syncope. This case is a good example of h significant and progressive condition.

Summary

- Aortic stenosis (AS) occurs when the aortic valve leaflets become stiff and are not able to fully open, making it difficult for the heart to pump blood out to the rest of the body.
- The three classic symptoms of AS are angina, syncope, and dyspnea.
- The three most common causes of AS are degenerative calcifications (leading to stenosis in elderly patients), congenital bicuspid valve (early stenosis in middle-aged patients), and rheumatic heart disease.
- The classic murmur associated with AS occurs in systole. The murmur decreases in intensity with a Valsalva maneuver or standing and increases with squatting.
- AS is associated with increased pressures to open the stiffened valve (afterload) and are susceptible to events which volume of blood in the LV. Thus, treatment with nitroglycerin, which causes venodilation and reduced preload should be avoided.
- The classic physical exam finding for AS is a systolic murmur that gets louder and then softer (crescendo-decrescendo); pulsus parvus et tardus is another.
- Echocardiograms are used to diagnose and grade the severity of AS.
- Patients with asymptomatic, severe AS can often be observed with serial echocardiograms.
- Symptomatic severe AS often requires aortic valve replacement.

Review Questions

1. A 45-year-old male born in the United States presents with shortness of breath and a feeling like he's going to pass out when he runs. Listening to his heart reveals a crescendo-decrescendo systolic murmur radiating to the carotids, and echocardiogram finds thickening and calcification (sclerosis) of the aortic valve. What else is most likely be found on echocardiogram?

A. ○ Aortic valve with one leaflet
B. ○ Aortic valve with three leaflets
C. ○ Aortic valve with three leaflets and mitral valve sclerosis
D. ● Aortic valve with two leaflets
E. ○ Vegetations on the aortic leaflets

Hide Explanation

The correct answer is aortic valve with two leaflets (D). This is a young patient with symptoms of AS (shortness of breath, lightheadedness, exercise intolerance). AS is not a very common diagnosis in middle-aged patients, so malformation of his aortic valve is likely, the most common of which is a bicuspid (two-leafed) valve. An aortic valve with one leaflet (A) is very rare, whereas an aortic valve with three leaflets (B) is a normal finding and would not help explain the patient's early presentation of AS. Aortic valve with three leaflets and mitral valve sclerosis (C) would suggest a history of rheumatic heart di tients from developed countries. Vegetations on the aortic leaflets (E) are suggestive of infective endocarditis, and this patient has no signs of infection.

2. You are listening to the heart sounds of a patient with severe AS. As you listen, you feel the carotid arteries in the patient's neck. What do you most likely detect?

A. ● Carotid pulses that happen after the heartbeat and feel weak
B. ○ Carotid pulses that happen before the heartbeat
C. ○ Carotid pulses that happen with the heartbeat
D. ○ Carotid pulses that happen with the heartbeat and have a humming feeling
E. ○ Carotid pulses that happen with the heartbeat and are very strong

Hide Explanation

The correct answer is carotid pulses that happen after the heartbeat and feel weak (A). This is called pulsus parvus et tardus and is a common finding with AS, especially if the AS is severe. Pulsus parvus et tardus is due to the heart's inability to pump much blood out of the left ventricle through the stenotic aortic valve. It's physically impossible to feel carotid pulses that happen before the heartbeat (B). Carotid pulses that happen with the heartbeat (C) are a normal finding and would not be expected in a patient with severe AS. Carotid pulses that happen with the heartbeat and have a AS and not as common as pulsus parvus et tardus. Carotid pulses that happen with the heartbeat and are very strong (E) can be a sign of the aortic valve not closing properly and the heart trying to compensate (aortic insufficiency), not AS.

3. A 78-year-old male with severe AS presents to the emergency room complaining of chest pain. Which of the following medications should not be prescribed?

A. ○ β-Blocker
B. ○ Calcium channel blocker
C. ○ Diuretic
D. ● Nitroglycerin

Hide Explanation

The correct answer is nitroglycerin (D), which decreases preload by causing vasodilation. In severe AS, patients are preload dependent, and reducing the preload with nitroglycerin (even though chest pain is typically treated with this medicine) could cause cardiovascular collapse. β-Blockers (A) or calcium channel blockers (B) might be appropriate if the patient has uncontrolled hypertension. Diuretics (C) could be appropriate if the patient has heart failure symptoms.

4. What findings ind acement?

A. ○ Mild AS on echocardiogram and leg swelling
B. ○ Moderate AS on echocardiogram, no symptoms, and no other heart problems
C. ○ Severe AS on echocardiogram and no symptoms at rest or on exercise stress testing
D. ● Severe AS on echocardiogram and shortness of breath at rest
E. ○ Severe AS on echocardiogram, no symptoms, and a fever of 103.0°F

Hide Explanation

The correct answer is severe AS on echocardiogram and shortness of breath at rest (D). Severe, symptomatic aortic stenosis (AS) is a clear indication for immediate aortic valve replacement (AVR). Mild AS on echocardiogram and leg swelling (A) is not an indication for immediate AVR because the heart failure (indicated by leg swelling) could be due to some other heart problem that should be addressed first. Moderate AS on echocardiogram, no symptoms, and no other heart problems (B) is not an indication for AVR. Patients with severe AS on echocardiogram and no symptoms at rest or on exercise stress testing (C) should only be operated on in specific circumstances, and, without more specifics, we would not operate on this patient. Severe AS on echocardiogram, no symptoms, and a fever of 103.0°F (E) is incorrect. If a patient is running a fever or has other signs of infection and has no symptoms from their AS (E), they should not undergo heart surgery until their infection is dealt with.

5. A patient with risk factors for aortic stenosis is also at risk for what other cardiac disease?

A. ○ Acute pericarditis
B. ● Coronary artery disease
C. ○ Dilated cardiomyopathy
D. ○ Endocarditis

Hide Explanation

The correct answer is coronary artery disease (B). The risk factors for aortic stenosis are the same as those for coronary artery disease (atherosclerosis), and include old age, male sex, smoking, hypertension, high LDL levels, and diabetes mellitus. The risk factors for acute pericarditis (A) include uremia, infections, autoimmune disease, and either recent myocardial infarction or heart surgery. The risk factors for dilated cardiomyopathy (C) include a family history, myocarditis, heavy alcohol consumption, or exposure to certain drugs. The risk factors for endocarditis (D) include bacteremia coupled with valvular abnormalities (i.e., infective endocarditis) or hypercoagulable conditions like malignancy or lupus (i.e., nonbacterial thrombotic endocarditis).

How would you rate this Brick?

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Acute Rheumatic Fever and Rheumatic Heart Disease

Last updated March 7, 2026 · 17 min read · ScholarRx

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Adapted from: Acute Rheumatic Fever and Rheumatic Heart Disease

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Define acute rheumatic fever (ARF) and rheumatic heart disease (RHD).
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